Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/13/2005
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Project Title:	Differential Gene Expression in Endothelial Cells Cultured in Rotating Wall Vessel Bioreactors & Exposed to Low Laminar Shear Stress		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-08-B
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HC81G		
Performance Goal No.:			
Performance Goal Text:			
	Rotating Wall Vessel Bioreactors offer a unique biotechnology approach to 3-D cell culture and advanced tissue engineering applications. In these fluid-filled Bioreactors, particles suspended in the culture medium rotate with the fluid in a quasi-solid body configuration. Cells and cell assemblies in this venue are in continual free fall and thus concomitantly exposed to low laminar shear stress (~ 0.5 dynes/cm2) and also to "simulated or modeled microgravity ", i.e. to a time-averaged, randomized gravitational force (~0.01 x g). To date, the contribution of each of these factors, especially of reduced laminar shear stress, has not been thoroughly assessed experimentally. Based on reports in the literature and our own preliminary data, we hypothesize that the component which simulates/models microgravity by "randomizing" the gravitational vector, rather than low, laminar flow–induced shear stress, may be instrumental in modulating gene expression and functional differentiation of cells/cell assemblies in RWV Bioreactors. In testing our		

Task Description:	hypothesis, we propose to use genomics, proteomics and bioinformatics approaches for analyzing patterns of differential expression of genes and gene products in human endothelial cells cultured for various amounts of time (4 h –20 days) in RWV Bioreactors, with controls maintained static monolayers, static suspension cultures, or as monolayers in a parallel plate flow chamber exposed to fluid shear stress of 0.5 dynes/cm2. In addition, by using the Perfused Rotating Wall Vessel Reactors, we will be able induce controlled levels of fluid shear stress levels in these Bioreactors. Endothelial cells are chosen for this study, since a) the fluid-shear dependent gene activation in these cells is well established and characterized b) altered fluid dynamics and mechanical forces, such as present in RWV Bioreactors (and space flight?) are known risk factors in the etiology of diminished cardiac and cardiovascular function, specifically in the etiology of atherosclerosis. The objectives of this study are as follows: Specific aim 1: Compare differential gene expression in human microvascular endothelial cells cultured respectively, as monolayers under static conditions, or exposed to low levels of steady shear stress, and in suspension culture in Rotating Wall Vessel Bioreactors. Specific Aim 2: Compare differential protein expression in human microvascular endothelial cells cultured under the same conditions described above in Specific Aim1. Specific aim 3: Use modern bioinformatics approaches for identifying functional correlations between expression patterns of genes and gene products, specifically of signaling molecules and pathways, which uniquely are expressed (only) in RWV Bioreactors Specific Aim 4: Use established molecular biological approaches for identifying a unique RWV –sensitive response element in the promoter region of one or more RWV-responsive genes. . While being responsive to several different stipulations of the soliciting NRA (01-OBPR-08), the objectives of this ground-based proposal is of relevance to NASA's mission, as outlined in the Critical Path Roadmap, specifically, Risk 6 Diminished Cardiac and Vascular Function. In this context our study adresses Enbling Questions 6a-6c: Does long-duration space flight lead to diminished cardiac function? What mechanisms are involved? Is the process reversible? • In studying the workings of NASA developed RWV Bioreactor Biotechnology we are focusing on a) elucidating the value of low shear culturing and b) studying differential endothelial gene expression in RWV Bioreactors. • This study will distinguish between the effects of low shear and that of other parameters in the workings of RWV Bioreactors, specifically those components that may represent a ground-based microgravity analog. Furthermore, the comprehensive approach will lead to the identification of candidate mechanism and genes/gene products that are uniquely responsive to the modeled microgravity component of the RWV environment. • The outcome of this study will facilitate rationally designing meaningful space flight experiments, aimed at deciphering the mechanisms by which cells and organisms sense and transduce microgravity. Thus, this study provides a strong link and a clear path to flight research on the International Space Station and beyond for long-duration space flight. • The proposed studies entail strong educational components, both for our students as well as for students from local/regional schools.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed study will enable the rational use of an important flight hardware, the RWV Bioreactor system, for ground based studies addressing important risks that figure prominetly in the Critical Road Map, such as such as cardiac and vascular functions, impaired immune responses, etc. In addition, the strong educational component of this study will be levaraged for outreach activities, to raise the level of awareness and enthusiasm for NASA-related projects amongst high school and college students.
Task Progress:	In the second year of this grant we have continued to make significant progress towards the first three of our four specific aims. According to our original timeline, we will begin tackling the fourth specific aim only in the last year of the grant. The initial analysis of our first genomics data sets is nearing completion. RNA for these data was obtained from 4 x 3 sets of human microvascular endothelial cells cultured respectively, as monolayers under static conditions, or exposed to low levels of steady shear stress, and in suspension culture in Rotating Wall Vessel Bioreactor (specific aim 1). At present we continue to mine our data in order to establish functional correlations between expression patterns of the genes which are differentially expressed under the various experimental conditions. A careful analysis of our initial data sets has revealed that amongst the ~ 20,000 human genes studied, more than 3000 are differentially expressed in one or more of the culture venues. For the quantitative analysis of these huge data sets, and for establishing functional correlations, we are using stae-of-the-art bioinformatics approaches and databases (such as the open-access data bases DAVID, KEGG, GO and, most recently, the commercial "gold-standard" Ingenuity). This level of sophisticated data mining will help us to elucidate some of the signaling molecules and pathways, which characterize the biological responses in each of the venues studied (specific aim3). Using these state-of-the art tools, we have identified a number of genes which the diverse venues have in common, but also others, which are uniquely expressed in each of the venues investigated. Importantly, we have identified a significant number of genes which seem to be uniquely are expressed (only) in RWV Bioreactors, and which may be related to the modeled microgravity component in this venue. As a validation of our approach, our data analysis of the date from the the low-shear environment has yielded a number of genes that are known to be activated by shear stress, for example some of the previously identified shear-responsive transcription factors. In addition, we identified some RWV-specific genes including a number of known genes previously associated with organogenesis. Our studies indicate potential novel functions for these genes, in terms of their being responsive to the RWV culture conditions. Importantly and in line with our hypothesis and our long-term goals, these genes might yield clues for developing general countermeasures for space-flight / microgravity associated impairments of human physiology. In addition to functioally classifying the many genes expressed in the various venues (gene-ontology), the use of that mining software, such as ingenuity, is helpful in identifying venue-specific central signaling pathways. This information, together with the knowledge on venue-specific genes described above, will be invaluable in identifying unique microgravity sensing/responsive similar mechanisms and in designing appropriate countermeasures. For the coming year, we anticipate to a) repeat the experiments with human microvascular cells for further validation of the complex data sets and for enhancing the statistical confidence in the results. Additional related experiments, planned as part of the goals for specific aim 4, involve establishing a causal relationship between the various venues and the observed venue-specific differential gene expression, by selectively down-regulating some of the unique, venue-specific signaling pathways, as identified in aim 1. In terms of specific aim 2, we have improved our capabilities to run 2-D gels so that we now can reliably resolve > 1000 proteins, as detailed in the previous progress report. The greatest challenge in the past year has been to increase the reproducibility of our 2-D gels. This was accomplished mainly by standardizing numerous technical issues, such as

	sample preparation, isolectrofocusing protocols, and 2-D gel running conditions. After establishing and validating the technical prerequisites for reproducible 2-D gel electrophoresis, we have now begun to compare differential protein expression in human endothelial cells cultured under the same conditions described above, using state-of-the art proteomics tools, such as DIGE (Differential-in-gel-electrophoresis) and Mass Spectrometry. Specifically, in these studies we aim at confirming at the protein level some of the lead candidates for differential gene expression, as described above under the progress in specific aim 1. In addition to the data mining results described above and in line with the goals for specific aim 3 indicated in last year's progress report, we have mainly focused in the past year on at on validating differential protein expression by "conventional" proteomics tools. In addition we are also using the same 2-D gel electrophoresis technology, a novel approach towards global assessment of protein phosphorylation - dependent signal transduction mechanisms. With this novel approach we are capitalizing on or past experience in analyzing phosphorylation -dependent signaling by 1-D gel electrophoresis. Using a phosphoprotein-specific stain, we have been able to identify dozens of proteins, which are differentially phosphorylated in the various venues. The focus of current and future efforts is to analyze the differential expression phosphoproteins, as a commensurate of cellular signaling. As before, we will use complex bioinformatics tools to elucidate differential activation of signaling pathways in the various venues. As before, knowledge of global signaling events common between the various venues and/or unique to the RWV environment, will allow us to design effective therapeutic targets to treat unique physiological disturbances, which will be useful for designee effective countermeasures at the molecular level. Finally, as a new technological advance, we are working on a novel optical system to determine in real-time the aggregation kinetics in RWV Bioreactors. The long-term goal of this engineering study is to use this information for a) obtaining detailed biological information from the parameters that affect the kinetics of aggregation and b) for developing feed-back algorithms, which will allow us to develop novel algorithms for automatically controlling the rotational speed and other vital parameters of RWV Bioreactors, based on the real-time monitoring of the aggregation process. We anticiapte that the novel methodology and algorithms developed in this work will enhance our ability for non-invasive, remote monitoring and data analysis during the exploration to Moon and Mars.
Bibliography Type:	Description: (Last Updated: 09/28/2023)
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